Education & Online Learning Pipeline

1) **Project Idea**

   Build a data pipeline that collects, processes, and analyzes online learning data to track:

   - Student performance
   - Course completion rates
   - Engagement levels
   - Dropout risks

   The system will help instructors and administrators make data-driven decisions.

2) **Data Sources**
   a) LMS Data base
   b) Platform Activity Logs

3) **Data types handled**
   a) **Structured Data**
      1) Student Id (Int)
      2) Grades (Float)
      3) Course IDs (String)
      4) Enrollment dates (Date)
   b) **Semi-Structured Data**
      1) JSON activity logs
   c) **Text Data**
      1) Student feedback comments
   d) **Timestamp Data**
      1) Login times
      2) Submission times

4) **Expected Challenges**
   a) Missing or incomplete student activity data

   b) Large log file processing

   c) Data privacy & security (student information)

   d) Data cleaning and normalization

   e) Real-time vs batch processing decisions

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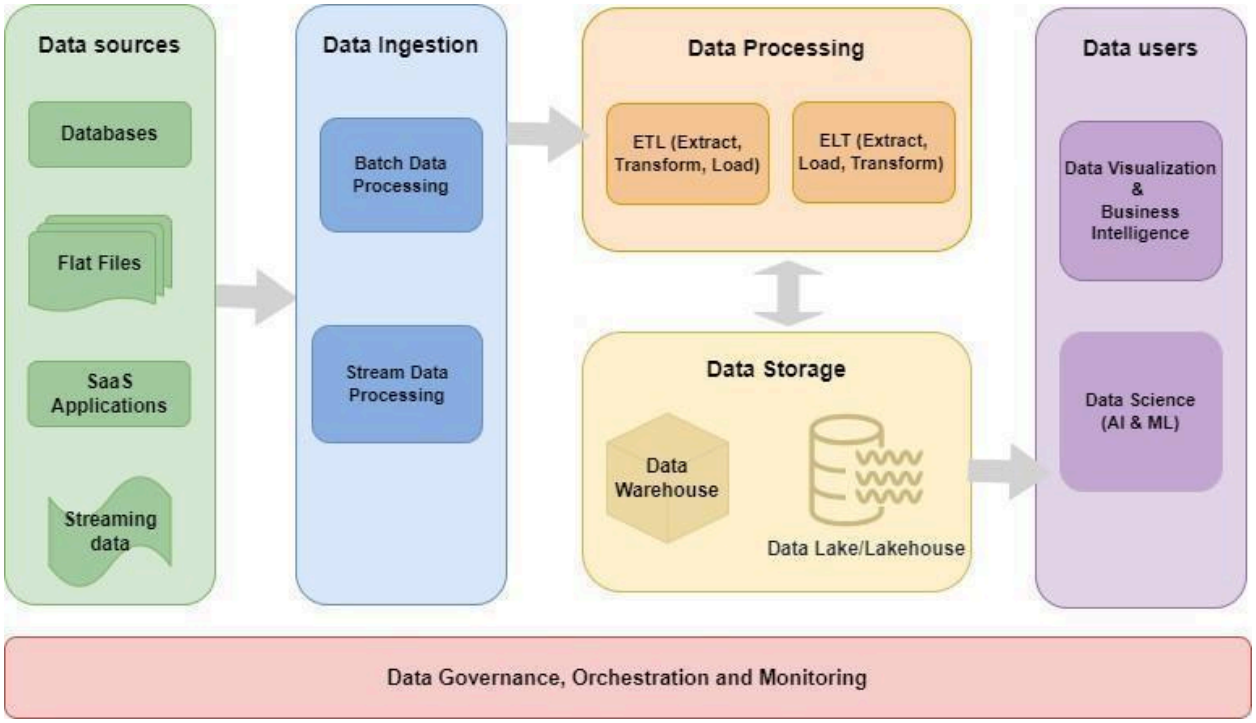
5) **Pipeline Architecture Diagram**



6) **Success Criteria**
   a) Accurate calculation of course completion rates
   b) Dashboard updates automatically
   c) Processing time under acceptable limit (e.g., < 5 minutes batch)
   d) Data accuracy ≥ 95%
   e) Student engagement metrics generated daily